ECE 252 Project – Phase Three

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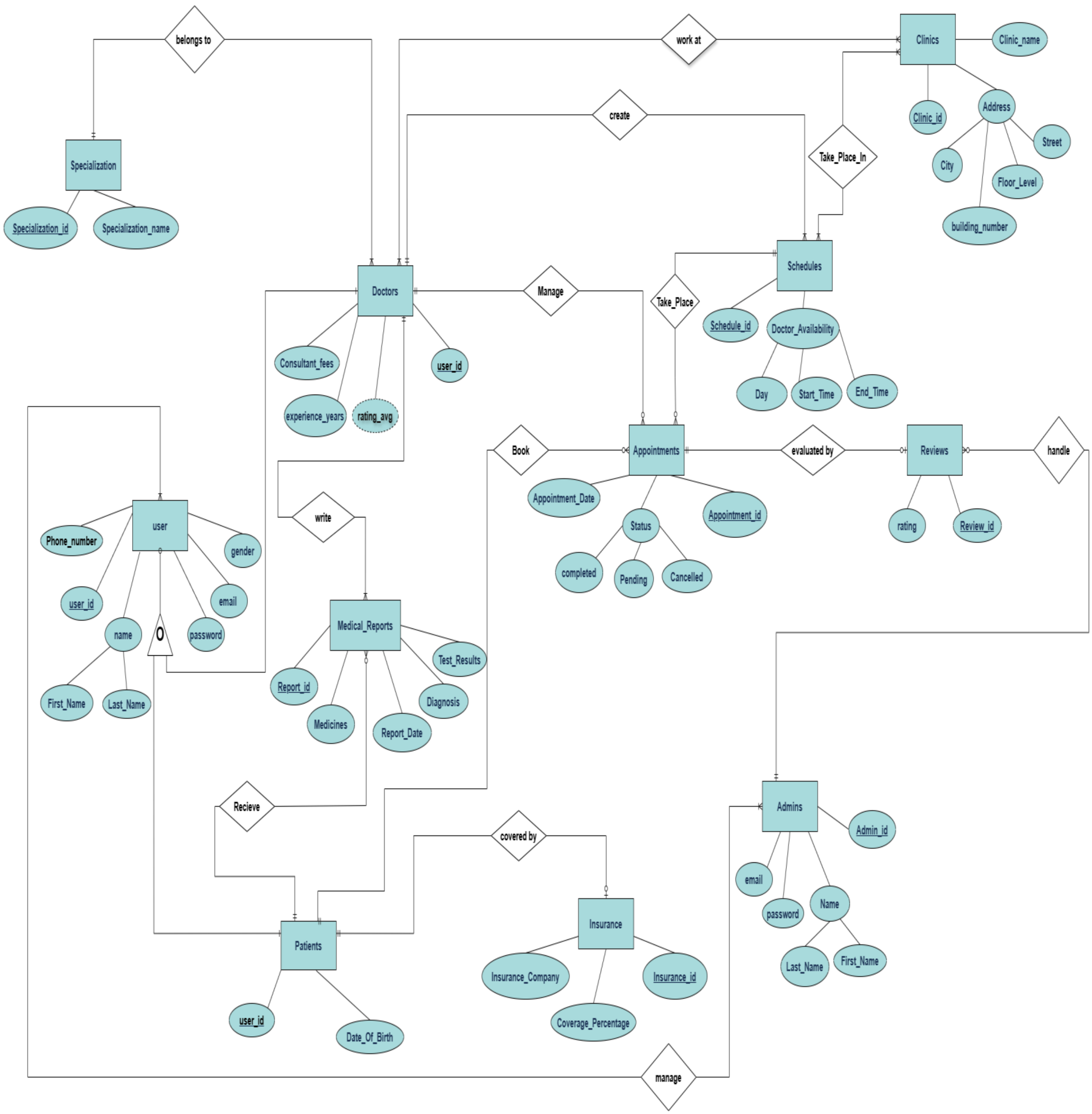
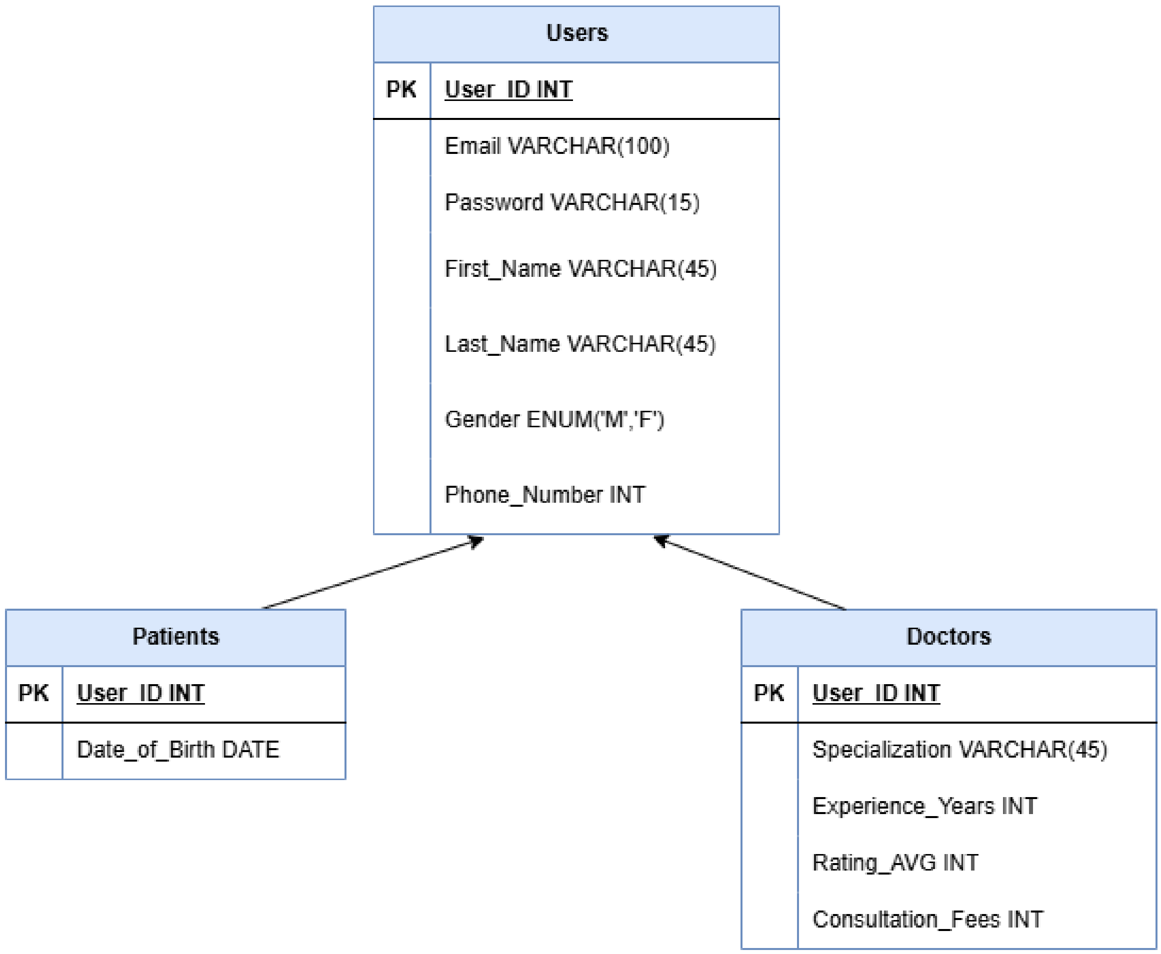
ER Diagram:-


belongs to
work at
Clinics
Clinic_name
Clinic_id
Address
Street
City
Floor_Level
building_number
create
Take_Place_In
Specialization
Specialization_id
Specialization_name
Doctors
Consultant_fees
experience_years
rating_avg
user_id
Manage
Take_Place
Schedules
Schedule_id
Doctor_Availability
Day
Start_Time
End_Time
Book
Appointments
Appointment_Date
Appointment_id
Status
completed
Pending
Cancelled
evaluated by
Reviews
rating
Review_id
handle
user
Phone_number
gender
user_id
email
name
password
First_Name
Last_Name
O
write
Medical_Reports
Report_id
Medicines
Report_Date
Diagnosis
Test_Results
Recieve
covered by
Insurance
Insurance_Company
Coverage_Percentage
Insurance_id
Patients
user_id
Date_Of_Birth
Admins
Admin_id
email
password
Name
Last_Name
First_Name
manage

Explanation of the ISA relationship between Patients & Doctors:



The ER diagram represents a **Smart Medical Appointment Booking System "E7gezly"** that includes several main **Entities**:

-The User entity is the "parent" entity which has User_ID (PK), Name, Gender, Phone number, Email, Password.

-The Doctor entity inherits the other attributes from "User" stores information such as User_ID (PK, FK), And others like consultation fees, and experience years, rating average and Clinic_ID (FK), Specialization_ID (FK), Schedule_ID (FK), Appointment_ID (FK), Report_ID (FK).

-The specialization entity has Specialization_ID (PK) and Specialization_Name and lastly UserDoctor_ID (FK)

-The Patient entity inherits the other attributes from "User". Patient entity includes User_ID (PK, FK), And others like date of birth and age, Report_ID (FK), Insurance_ID (FK), Appointment_ID (FK).

-The Appointment entity represents the booking between a patient and a doctor and contains Appointment_ID (PK), date, time, status, along with UserPatient_ID (FK) and UserDoctor_ID (FK) to link each appointment to a specific patient and doctor. Reviews_ID (FK), Schedules_ID (FK).

-The Review entity includes Review_ID (PK), rating and uses Admins_ID (FK) and Appointments_ID (FK) to represent that a patient writes a review for a doctor.

-The Medical_Report entity stores medical details such as Report_ID (PK), diagnosis, treatment, date, tests and includes UserPatient_ID (FK) UserDoctor_ID (FK) to associate reports with patients.

-The Schedule entity contains Schedule_ID (PK), available days and times, and includes UserDoctor_ID (FK) to indicate which doctor the schedule belongs to, Clinic_ID (FK) and Appointments_ID (FK).

-The Clinic entity includes Clinic_ID (PK), Name and location also Schedules_ID (FK) and UserDoctors_ID (FK).

-The Insurance entity includes Insurance_ID (PK), company name, and coverage details, and is linked to patients through UserPatient_ID (FK).

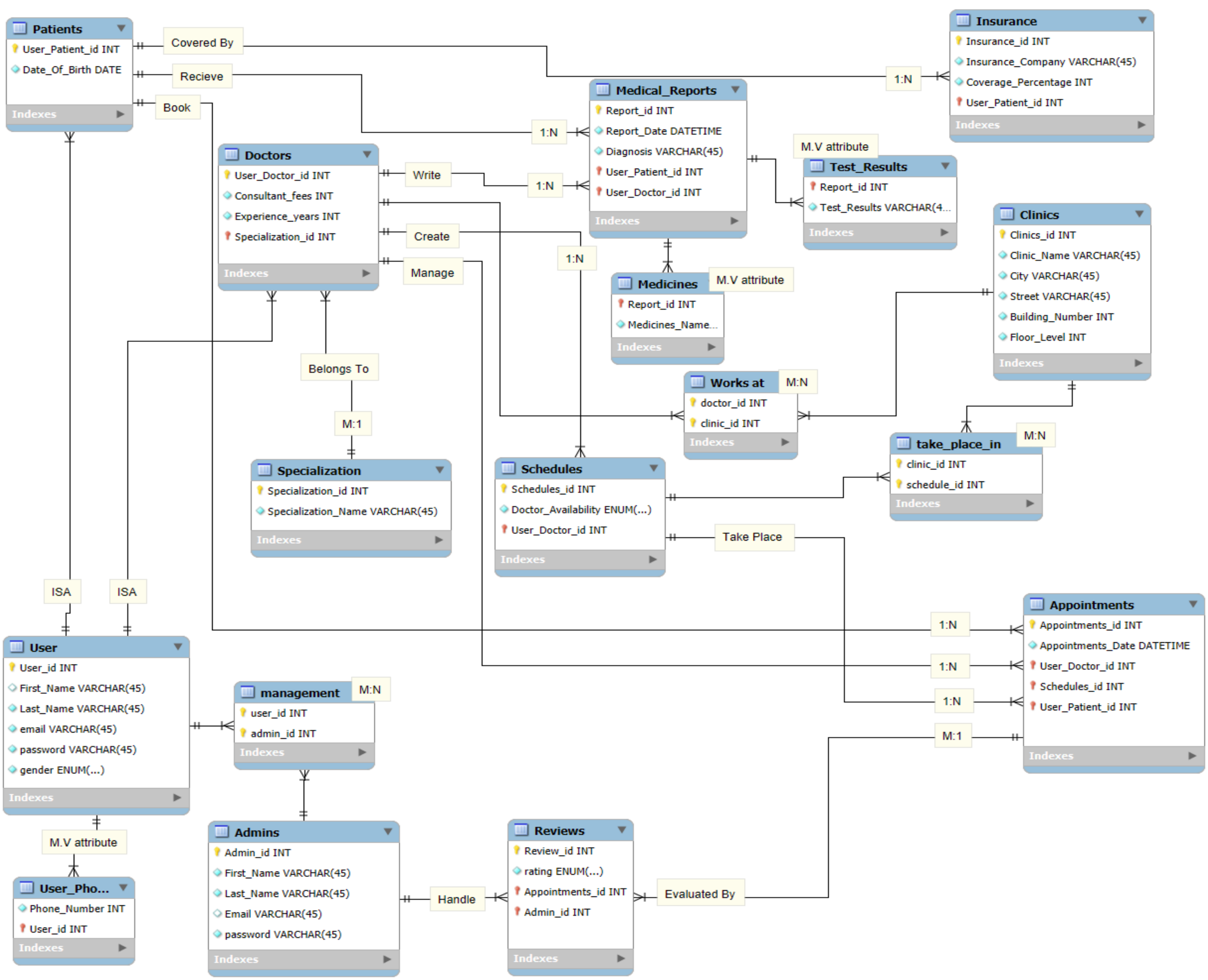
-Finally, the Admin entity contains Admin_ID (PK), Name, Password and mail, Also Reviews_ID (FK) and User_ID (FK).

Relationships & 1NF:

The relationships between these entities show that a patient can book multiple appointments, a doctor can handle many appointments, patients can write reviews for doctors only when the appointment is completed, and each doctor operates within a clinic or more and follows a schedule, and there are Admins who handle reviews and manage users, forming a complete and structured healthcare system.

And all the participations of the relationships are "mandatory".

Regarding normalization, the database satisfies First Normal Form (1NF) because all attributes in each entity are atomic, meaning each field contains only a single value and there are no repeating groups or multi-valued attributes. For example, instead of storing multiple appointments in one field, each appointment is stored as a separate record in the Appointment table, and each attribute (such as date, time, or status) holds only one value. This ensures that the data is going to be organized in a structured tabular form with clear primary keys and no duplicated or nested data within a single column.



2NF & 3NF:

Moving from 1NF to 2NF in this schema eliminates all partial dependencies where non-key attributes were depending on only *part* of a composite primary key rather than the whole key. However, it is worth noting that all tables in the schema already rely on single-column primary keys, which by definition cannot have partial dependencies since there is no composite key to be partially dependent on, meaning the structural difference between 1NF and 2NF here is relatively contained, with the junction tables being the primary area of concern rather than the schema as a whole.

Moving from 2NF to 3NF in this schema eliminates all transitive dependencies, where non-key attributes were depending on other non-key attributes rather than directly on the primary key. Three main changes occur across the schema. First, rating_avg is dropped from the Doctors table entirely, since it is just a calculated average derivable from the Reviews table at any time, meaning storing it directly was creating a risk of stale or inconsistent data. Second, Coverage_Percentage is extracted out of the Insurance table and moved into a dedicated Insurance_Company table, because the coverage percentage is really a property of the company itself rather than each individual insurance record, so if a company changes its rate, one update now covers all patients instead of dozens of rows. Third, City is separated from the Clinics table into a dedicated Location table alongside Street, since a city is technically a property of a street rather than an independent fact about the clinic itself, eliminating the risk of the same street being accidentally assigned to two different cities across different records. Together these changes ensure that every attribute on every table describes only and directly its primary key, producing a cleaner, more consistent schema where redundancy is minimized and updates never risk.

Code MySQL:

```
CREATE DATABASE e7gezly;

USE e7gezly;


CREATE TABLE Users (

    User_id INT PRIMARY KEY AUTO_INCREMENT,

    First_Name VARCHAR(45) NOT NULL,

    Last_Name VARCHAR(45) NOT NULL,

    Email VARCHAR(100) NOT NULL UNIQUE,
```

```
    Password VARCHAR(255) NOT NULL,
    Gender ENUM('Male','Female') NOT NULL
);


CREATE TABLE User_Phone (
    User_id INT,
    Phone_Number INT ,

    PRIMARY KEY (User_id, Phone_Number),

    FOREIGN KEY (User_id)
    REFERENCES Users(User_id)
);


CREATE TABLE Patients (
    User_Patient_id INT PRIMARY KEY,
    Date_Of_Birth DATE NOT NULL,

    FOREIGN KEY (User_Patient_id)
    REFERENCES Users(User_id)
);



CREATE TABLE Specialization (
    Specialization_id INT PRIMARY KEY AUTO_INCREMENT,
    Specialization_Name VARCHAR(45) NOT NULL UNIQUE
```

```
);


CREATE TABLE Doctors (
    User_Doctor_id INT PRIMARY KEY,
    Consultant_Fees INT NOT NULL,
    Experience_Years INT NOT NULL,
    Specialization_id INT NOT NULL,

    FOREIGN KEY (User_Doctor_id)
    REFERENCES Users(User_id),

    FOREIGN KEY (Specialization_id)
    REFERENCES Specialization(Specialization_id)
);


CREATE TABLE Admins (
    Admin_id INT PRIMARY KEY AUTO_INCREMENT,
    First_Name VARCHAR(45) NOT NULL,
    Last_Name VARCHAR(45) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Password VARCHAR(255) NOT NULL
);


CREATE TABLE Management (
    User_id INT,
    Admin_id INT,
```

```
    PRIMARY KEY (User_id, Admin_id),

    FOREIGN KEY (User_id)
    REFERENCES Users(User_id),

    FOREIGN KEY (Admin_id)
    REFERENCES Admins(Admin_id)
);

CREATE TABLE Insurance_Company (
    Company_id INT PRIMARY KEY AUTO_INCREMENT,
    Company_Name VARCHAR(45) NOT NULL,
    Coverage_Percentage INT CHECK (Coverage_Percentage BETWEEN 0 AND 100)
);


CREATE TABLE Insurance (
    Insurance_id INT PRIMARY KEY AUTO_INCREMENT,
    Company_id INT NOT NULL,
    User_Patient_id INT NOT NULL,

    FOREIGN KEY (Company_id)
    REFERENCES Insurance_Company(Company_id),

    FOREIGN KEY (User_Patient_id)
    REFERENCES Patients(User_Patient_id)
);
```

```
CREATE TABLE Location (
    Location_id INT PRIMARY KEY AUTO_INCREMENT,
    Street VARCHAR(45) NOT NULL,
    City VARCHAR(45) NOT NULL
);


CREATE TABLE Clinics (
    Clinic_id INT PRIMARY KEY AUTO_INCREMENT,
    Clinic_Name VARCHAR(45) NOT NULL,
    Location_id INT NOT NULL,
    Building_Number INT NOT NULL,
    Floor_Level INT NOT NULL,

    FOREIGN KEY (Location_id)
    REFERENCES Location(Location_id)
);


CREATE TABLE Works_At (
    Doctor_id INT,
    Clinic_id INT,

    PRIMARY KEY (Doctor_id, Clinic_id),

    FOREIGN KEY (Doctor_id)
    REFERENCES Doctors(User_Doctor_id),
```

```sql
    FOREIGN KEY (Clinic_id)
    REFERENCES Clinics(Clinic_id)
);


CREATE TABLE Schedules (
    Schedule_id INT PRIMARY KEY AUTO_INCREMENT,
    Day_Name VARCHAR(20) NOT NULL,
    Start_Time TIME NOT NULL,
    End_Time TIME NOT NULL,
    User_Doctor_id INT NOT NULL,

    FOREIGN KEY (User_Doctor_id)
    REFERENCES Doctors(User_Doctor_id)
);

CREATE TABLE Take_Place_In (
    Clinic_id INT,
    Schedule_id INT,

    PRIMARY KEY (Clinic_id, Schedule_id),

    FOREIGN KEY (Clinic_id)
    REFERENCES Clinics(Clinic_id),

    FOREIGN KEY (Schedule_id)
    REFERENCES Schedules(Schedule_id)
);
```

```
CREATE TABLE Appointments (
    Appointment_id INT PRIMARY KEY AUTO_INCREMENT,
    Appointment_Date DATETIME NOT NULL,
    User_Doctor_id INT NOT NULL,
    Schedule_id INT NOT NULL,
    User_Patient_id INT NOT NULL,

    FOREIGN KEY (User_Doctor_id)
    REFERENCES Doctors(User_Doctor_id),

    FOREIGN KEY (Schedule_id)
    REFERENCES Schedules(Schedule_id),

    FOREIGN KEY (User_Patient_id)
    REFERENCES Patients(User_Patient_id)
);

CREATE TABLE Medical_Reports (
    Report_id INT PRIMARY KEY AUTO_INCREMENT,
    Report_Date DATETIME NOT NULL,
    Diagnosis VARCHAR(100) NOT NULL,
    User_Patient_id INT NOT NULL,
    User_Doctor_id INT NOT NULL,

    FOREIGN KEY (User_Patient_id)
    REFERENCES Patients(User_Patient_id),

    FOREIGN KEY (User_Doctor_id)
```

```sql
    REFERENCES Doctors(User_Doctor_id)
);


CREATE TABLE Medicines (
    Report_id INT,
    Medicine_Name VARCHAR(45),

    PRIMARY KEY (Report_id, Medicine_Name),

    FOREIGN KEY (Report_id)
    REFERENCES Medical_Reports(Report_id)
);


CREATE TABLE Test_Results (
    Report_id INT,
    Test_Result VARCHAR(45),

    PRIMARY KEY (Report_id, Test_Result),

    FOREIGN KEY (Report_id)
    REFERENCES Medical_Reports(Report_id)
);


CREATE TABLE Reviews (
    Review_id INT PRIMARY KEY AUTO_INCREMENT,
    Rating TINYINT CHECK (Rating BETWEEN 1 AND 5),
```

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    Appointment_id INT NOT NULL,
    Admin_id INT NOT NULL,

    FOREIGN KEY (Appointment_id)
    REFERENCES Appointments(Appointment_id),

    FOREIGN KEY (Admin_id)
    REFERENCES Admins(Admin_id)
);

INSERT INTO Users (First_Name, Last_Name, Email, Password, Gender) VALUES
('Ahmed','Hassan','ahmed.hassan@gmail.com','ah123','Male'),
('Sara','Ali','sara.ali@gmail.com','sa123','Female'),
('Mohamed','Adel','mohamed.adel@gmail.com','mo123','Male'),
('Nour','Khaled','nour.khaled@gmail.com','no123','Female'),
('Omar','Samy','omar.samy@gmail.com','om123','Male'),
('Mariam','Yasser','mariam.yasser@gmail.com','ma123','Female'),
('Youssef','Nabil','youssef.nabil@gmail.com','yo123','Male'),
('Salma','Tarek','salma.tarek@gmail.com','st123','Female'),
('Karim','Fathy','karim.fathy@gmail.com','ka123','Male'),
('Laila','Mostafa','laila.mostafa@gmail.com','la123','Female'),
('Hany','Ibrahim','hany.ibrahim@gmail.com','ha123','Male'),
('Dina','Maher','dina.maher@gmail.com','di123','Female'),
('Amr','Gamal','dr.amr.gamal@e7gezly.com','am123','Male'),
('Reem','Ashraf','dr.reem.ashraf@e7gezly.com','re123','Female'),
('Tamer','Lotfy','dr.tamer.lotfy@e7gezly.com','ta123','Male'),
('Farah','Wael','dr.farah.wael@e7gezly.com','fa123','Female'),
('Khaled','Ragab','dr.khaled.ragab@e7gezly.com','kh123','Male'),
('Menna','Sherif','dr.menna.sherif@e7gezly.com','me123','Female'),
```

```sql
('Ali','Fouad','dr.ali.fouad@e7gezly.com','al123','Male'),
('Habiba','Ehab','dr.habiba.ehab@e7gezly.com','hb123','Female');


INSERT INTO User_Phone VALUES
(1,'01011111111'),
(2,'01022222222'),
(3,'01033333333'),
(4,'01044444444'),
(5,'01055555555'),
(6,'01066666666'),
(7,'01077777777'),
(8,'01088888888'),
(9,'01099999999'),
(10,'01100000000'),
(11,'01111111111'),
(12,'01122222222'),
(13,'01133333333'),
(14,'01144444444'),
(15,'01155555555'),
(16,'01166666666'),
(17,'01177777777'),
(18,'01188888888'),
(19,'01199999999'),
(20,'01200000000');


INSERT INTO Patients VALUES
(1,'1999-05-10'),
```

```sql
(2,'2001-03-15'),
(3,'1998-07-21'),
(4,'2000-11-30'),
(5,'1997-01-25'),
(6,'2002-08-18'),
(7,'1995-12-05'),
(8,'2003-06-09'),
(9,'1996-04-14'),
(10,'2001-09-27'),
(11,'1994-02-11'),
(12,'2000-10-19');


INSERT INTO Specialization (Specialization_Name) VALUES
('Cardiology'),
('Dermatology'),
('Neurology'),
('Pediatrics'),
('Orthopedics');


INSERT INTO Doctors VALUES
(13,500,10,1),
(14,400,7,2),
(15,650,12,3),
(16,300,5,4),
(17,550,9,5),
(18,450,6,1),
(19,350,4,2),
```

```sql
(20,700,15,3);

INSERT INTO Admins (First_Name, Last_Name, Email, Password) VALUES
('Admin','One','admin1@e7gzly.com','admin123'),
('Admin','Two','admin2@e7gzly.com','admin456'),
('Admin','Three','admin3@e7gzly.com','admin789');

INSERT INTO Management VALUES
(1,1),
(2,1),
(13,2),
(14,2),
(15,3);

INSERT INTO Insurance_Company (Company_Name, Coverage_Percentage) VALUES
('AXA',80),
('MetLife',70),
('Allianz',85),
('MedNet',75),
('NextCare',90);

INSERT INTO Insurance (Company_id, User_Patient_id) VALUES
(1,1),
(2,2),
(3,3),
(4,4),
(5,5),
```

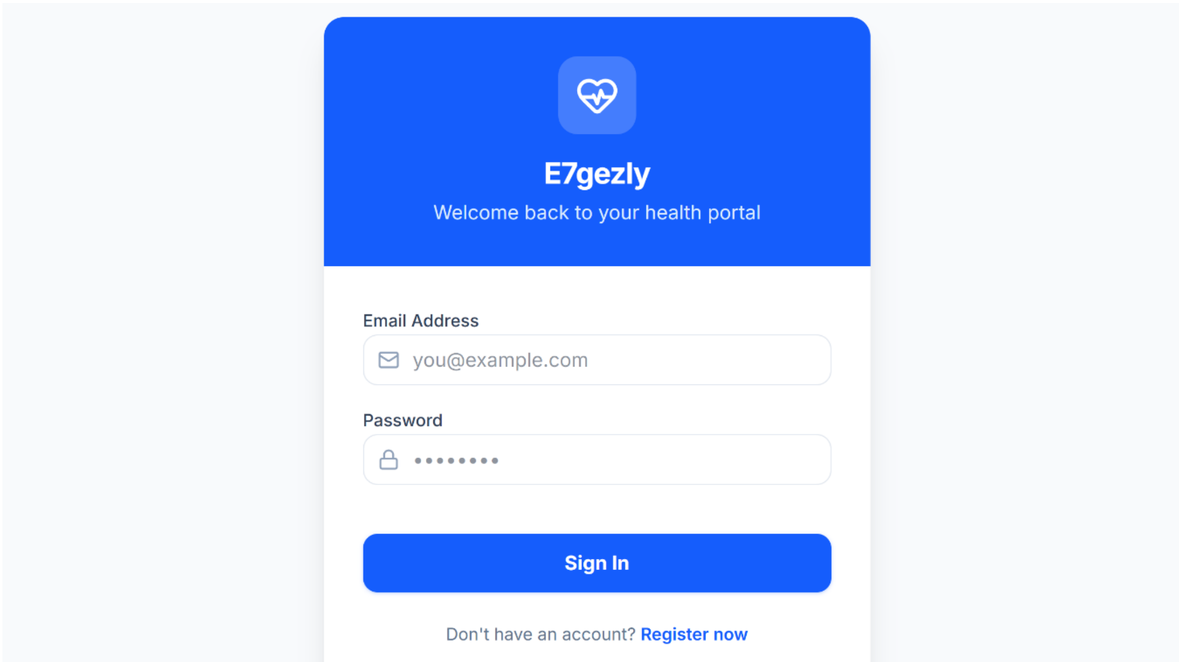
```sql
(1,6),
(2,7),
(3,8),
(4,9),
(5,10);


INSERT INTO Location (Street, City) VALUES
('Tahrir Street','Cairo'),
('Nasr Road','Cairo'),
('Corniche','Alexandria'),
('University Street','Mansoura'),
('Sea Road','Hurghada');

INSERT INTO Clinics
(Clinic_Name, Location_id, Building_Number, Floor_Level)
VALUES
('Heart Care Clinic',1,10,2),
('Skin Health Center',2,15,3),
('Brain Clinic',3,20,1),
('Kids Clinic',4,8,2),
('Bone Care Clinic',5,12,4);

INSERT INTO Works_At VALUES
(13,1),
(14,2),
(15,3),
(16,4),
(17,5),
```

```
(18,1),
(19,2),
(20,3);

INSERT INTO Schedules
(Day_Name, Start_Time, End_Time, User_Doctor_id)
VALUES
('Sunday','09:00:00','01:00:00',13),
('Monday','10:00:00','02:00:00',14),
('Tuesday','11:00:00','03:00:00',15),
('Wednesday','08:00:00','12:00:00',16),
('Thursday','01:00:00','05:00:00',17),
('Friday','09:00:00','01:00:00',18),
('Saturday','10:00:00','02:00:00',19),
('Sunday','12:00:00','04:00:00',20);

INSERT INTO Take_Place_In VALUES
(1,1),
(2,2),
(3,3),
(4,4),
(5,5),
(1,6),
(2,7),
(3,8);
```

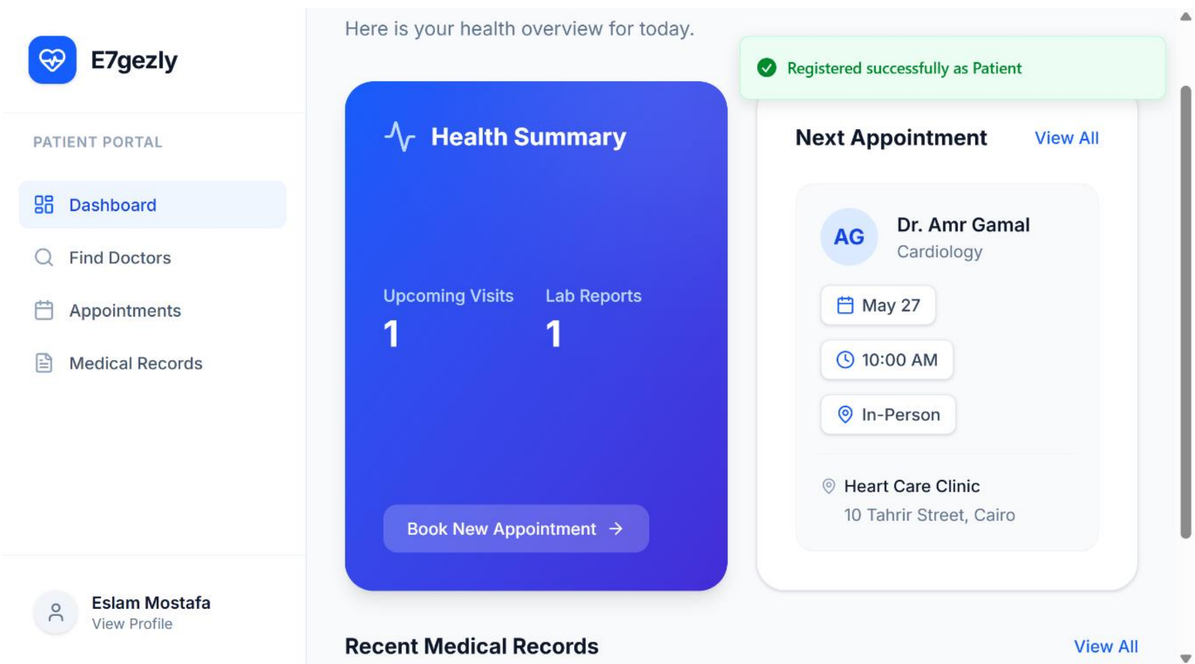
```sql
INSERT INTO Appointments
(Appointment_Date, User_Doctor_id, Schedule_id, User_Patient_id)
VALUES
('2026-05-20 10:00:00',13,1,1),
('2026-05-20 11:00:00',14,2,2),
('2026-05-21 12:00:00',16,4,3),
('2026-05-22 01:00:00',17,5,4),
('2026-05-23 02:00:00',19,7,5),
('2026-05-24 03:00:00',20,8,6);


INSERT INTO Medical_Reports
(Report_Date, Diagnosis, User_Patient_id, User_Doctor_id)
VALUES
('2026-05-20 10:30:00','High Blood Pressure',1,13),
('2026-05-20 11:30:00','Skin Allergy',2,14),
('2026-05-21 12:30:00','Flu',3,16),
('2026-05-22 01:30:00','Knee Pain',4,17),
('2026-05-23 02:30:00','Migraine',5,20);


INSERT INTO Medicines VALUES
(1,'Panadol'),
(1,'Aspirin'),
(2,'Cetirizine'),
(3,'Paracetamol'),
(4,'Ibuprofen'),
(5,'Migranil');
```

```
INSERT INTO Test_Results VALUES
(1,'Blood Pressure Test'),
(2,'Skin Test'),
(3,'CBC'),
(4,'X-Ray'),
(5,'MRI');


INSERT INTO Reviews
(Rating, Appointment_id, Admin_id)
VALUES
(5,1,1),
(4,2,2),
(3,3,1),
(5,4,3),
(4,5,2),
(5,6,3);


SELECT
   User_Doctor_id,
   AVG(Rating) AS Rating_Average
FROM Reviews
JOIN Appointments
ON Reviews.Appointment_id = Appointments.Appointment_id
GROUP BY User_Doctor_id;
```

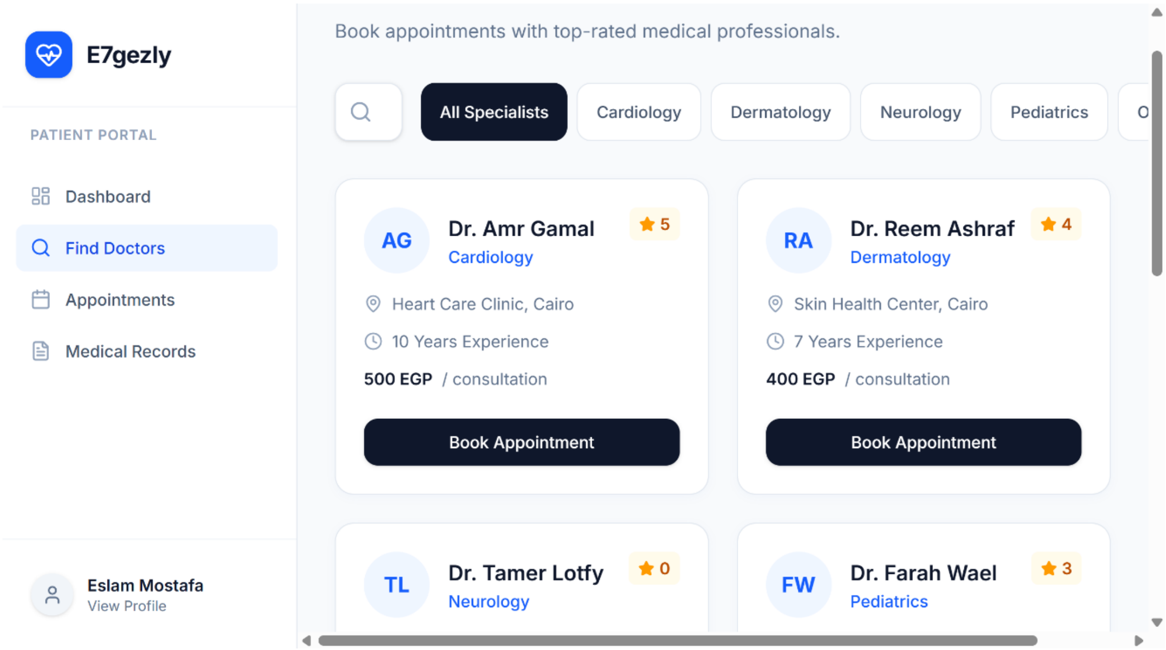
- **Login interface:**



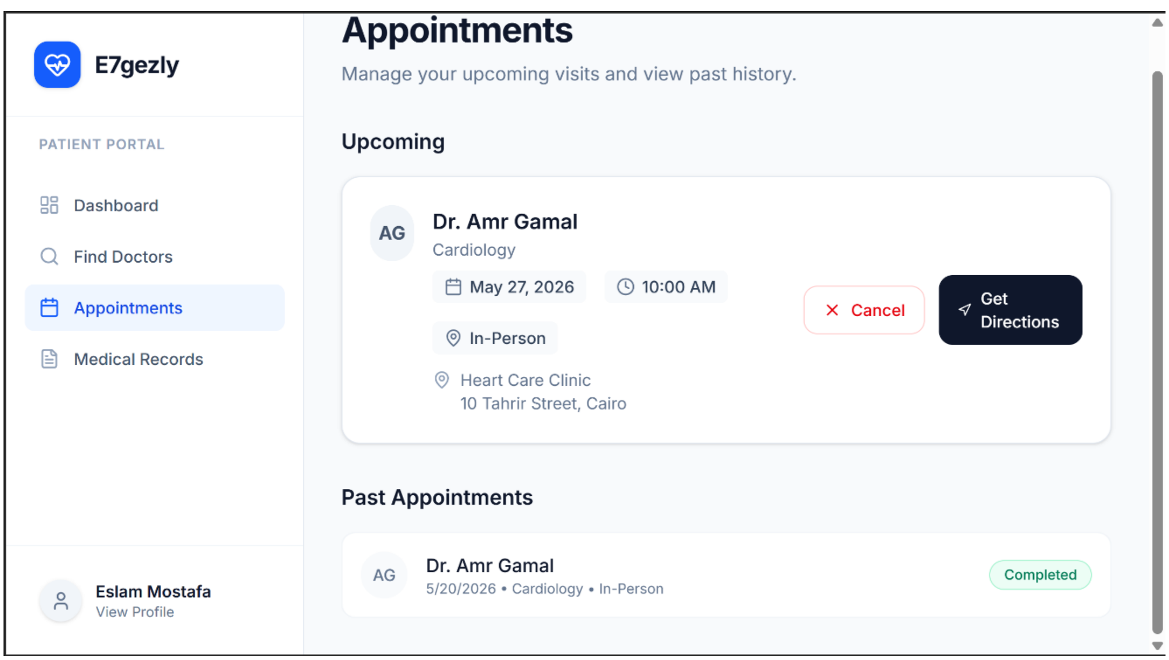
- **Account Created:**



➢ Booking Appointment:



➢ Appointment Booked:



➢ Appointment Cancelled: